

Collecting Like Terms

Example

Simplify: $4a + 8b - 2a - 5b$

What do you think the answer will be to this question?

Perhaps $6a - 3b$? How about $2a - 3b$?

Unfortunately, these answers are incorrect!

Here's how to simplify an expression:

Step 1:

Find two different colours (or however many different 'endings' you have to each term). Here we only have a's and b's so we only need 2 colours.

Step 2:

Circle and collect all the **a**s in one colour. It is important that you circle the sign to the left of it as well, as this will tell you whether you need to add or subtract the like terms.

$$\begin{array}{ccccccc} 4a & + & 8b & - & 2a & - & 5b \\ & & & & \swarrow & & \swarrow \\ & & & & 4a & - & 2a \\ & & & & 2a & & \end{array}$$

Step 3:

Circle and collect the **b**s in the other colour.

$$\begin{array}{ccccccc} 4a & + & 8b & - & 2a & - & 5b \\ & & \swarrow & & \swarrow & & \swarrow \\ & & + 8b & - & 5b & & \\ & & + 3b & & & & \end{array}$$

Step 4:

Put everything together:

Your final answer is $2a + 3b$

$$\begin{array}{ccccccc} 4a & + & 8b & - & 2a & - & 5b \\ \swarrow & & \swarrow & & \swarrow & & \swarrow \\ 4a & - & 2a & & + & 8b & - & 5b \\ 2a & & & & + & 3b & & \end{array}$$

Example

Simplify: $2x^2 + 4x + 12 - 2x$

Step 1:

There are 3 different 'endings' to the terms here so you will need 3 different colours. Remember x^2 is a different 'ending' to just x .

Step 2:

Circle $2x^2$ in one colour.

$$\begin{array}{ccccccc} 2x^2 & + & 4x & + & 12 & - & 2x \\ & & & & \swarrow & & \swarrow \\ & & & & 2x^2 & & \end{array}$$

Step 3:

Circle and collect the **x**s in another colour. Don't forget to include the signs to the left of them.

$$\begin{array}{ccccccc} 2x^2 & + & 4x & + & 12 & - & 2x \\ & & \swarrow & & \swarrow & & \swarrow \\ & & + 4x & - & 2x & & \\ & & + 2x & & & & \end{array}$$

Step 4:

Circle the 12 in another colour.

$$\begin{array}{ccccccc} 2x^2 & + & 4x & + & 12 & - & 2x \\ & & & & \swarrow & & \swarrow \\ & & & & + 12 & & \end{array}$$

Step 5:

Put everything together:

Your final answer is $2x^2 + 2x + 12$